INCH-POUND

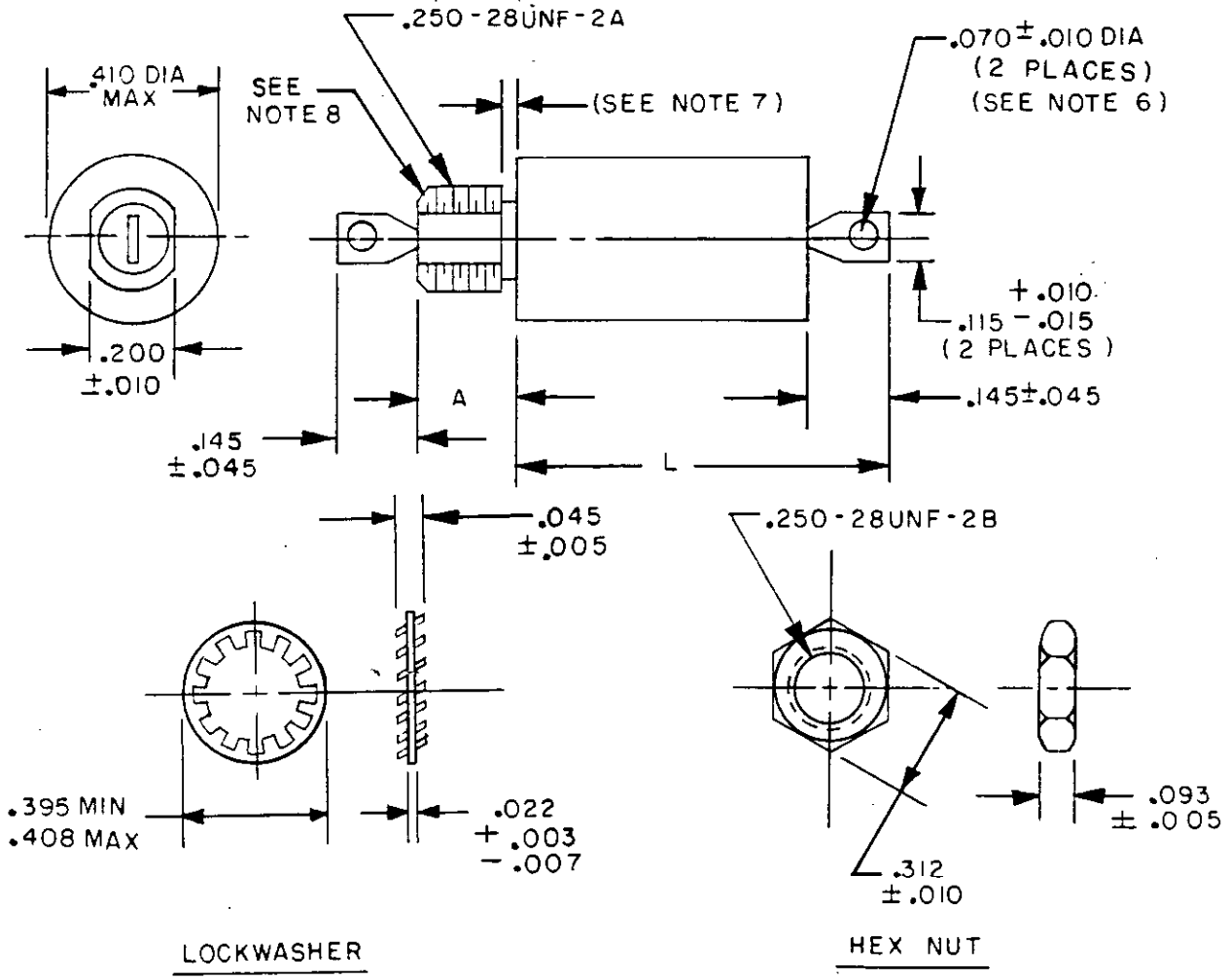
MIL-F-28861/4A
2 November 1988
SUPERSEDING
MIL-F-28861/4
17 December 1981

MILITARY SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLE FS40

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.



MOUNTING HARDWARE

FIGURE 1. Case and hardware dimensions and circuit diagrams.

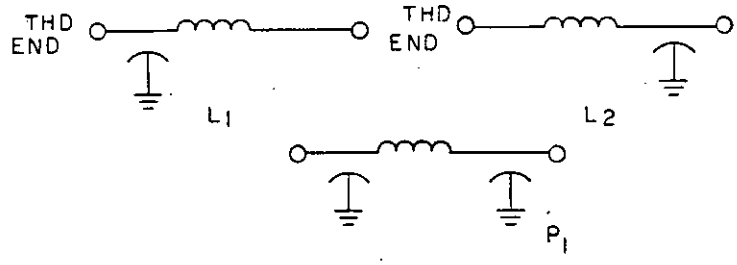
(A) Denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Dash number	A dimension (±.010)	L dimension max	Dash number	A dimension (±.010)	L dimension max
001	.187	.630	019	.312	.630
002	.187	.630	020	.312	.630
003	.187	.730	021	.312	.730
004	.187	.630	022	.312	.630
005	.187	.630	023	.312	.630
006	.187	.730	024	.312	.730
007	.187	.630	025	.312	.630
008	.187	.630	026	.312	.630
009	.187	.730	027	.312	.730
010	.187	.630	028	.312	.630
011	.187	.630	029	.312	.630
012	.187	.730	030	.312	.730
013	.187	.630	031	.312	.630
014	.187	.630	032	.312	.630
015	.187	.730	033	.312	.730
016	.187	.630	034	.312	.630
017	.187	.630	035	.312	.630
018	.187	.730	036	.312	.730

Inches	mm	Inches	mm
.003	0.08	.145	3.68
.005	0.13	.187	4.75
.007	0.18	.200	5.08
.010	0.25	.250	6.35
.015	0.38	.312	7.92
.022	0.56	.395	10.03
.045	1.14	.408	10.36
.070	1.78	.410	10.41
.093	2.36	.630	16.00
.115	2.92	.730	18.54



CIRCUIT DIAGRAMS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter with the symbol "C" or the symbol "L", as follows, or the circuit diagram shall be marked on the case.

Circuit	Symbol
L_1	C
L_2	L

6. Optional slot may be supplied, .050 ±.010 inches (1.27 ±0.25 mm) x .070 ±.010 inches (1.78 ±0.25 mm).
7. Imperfect thread or undercut optional .050 inch (1.27 mm) maximum.
8. One imperfect thread allowed .035 inch (0.89 mm) maximum.
9. Recommended mounting torque: 44 oz-in ±4 oz-in.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 10 grams maximum.

Case finish:

Class B filters, T only (tin plated or tin-lead plated).

(A) Classes A and S filters, T or G only (tin plated, tin-lead plated, or gold plated). In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: 70 V dc.

Rated current: See table I.

Capacitance: See table I for minimum capacitance value.

Dissipation factor: 3 percent maximum.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohms minimum or 1,000 megohm-microfarads minimum, whichever is less.

At +125°C: 100 megohms minimum or 100 megohm-microfarads minimum, whichever is less.

Insertion loss: See table I.

Voltage drop: See table I.

DC resistance: See table I.

(A) Seal: In accordance with MIL-F-28861. Leakage rate for classes A and S filters shall not exceed 1×10^{-7} atm cm^3/s.

Temperature rise: +25°C maximum.

Solderability of terminals: In accordance with MIL-F-28861.

(A) Quality assurance provisions:

Class B and class S filters: In accordance with MIL-F-28861.

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Ⓐ TABLE I. Electrical characteristics.

Dash number	Product assurance level			Circuit	Maximum current (amps)	Minimum capacitance (μF)	Maximum voltage drop (volts)	Maximum DC resistance (ohms)	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/																			
									At +25°C												At -55°C and +125°C										
	Class								15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz
	A	B	S																									
001, 019	X	X	X	L1	0.10	0.70	0.17	1.7	9	20	29	41	48	60	70	70	70	70	7	18	27	39	46	58	68	70	70	70
002, 020	X	X	X	L2	0.10	0.70	0.17	1.7	9	20	29	41	48	60	70	70	70	70	7	18	27	39	46	58	68	70	70	70
003, 021	X	X	X	π	0.10	1.4	0.17	1.7	15	36	50	69	79	80	80	80	80	80	13	34	48	67	77	78	78	80	80	80
004, 022	X	X	X	L1	0.30	0.70	0.23	0.77	6	15	23	35	42	54	70	70	70	70	4	13	21	33	40	68	68	70	70	70
005, 023	X	X	X	L2	0.30	0.70	0.23	0.77	6	15	23	35	42	54	70	70	70	70	4	13	21	33	40	68	68	70	70	70
006, 024	X	X	X	π	0.30	1.4	0.23	0.77	---	29	44	62	73	80	80	80	80	80	---	27	42	60	71	78	78	80	80	80
007, 025	X	X	X	L1	0.50	0.70	0.18	0.36	5	12	19	29	36	48	69	70	70	70	3	10	17	27	34	46	67	70	70	70
008, 026	X	X	X	L2	0.50	0.70	0.18	0.36	5	12	19	29	36	48	69	70	70	70	3	10	17	27	34	46	67	70	70	70
009, 027	X	X	X	π	0.50	1.4	0.18	0.36	---	21	37	56	67	80	80	80	80	80	---	19	35	54	65	78	78	80	80	80
010, 028	X	X	X	L1	1.0	0.70	0.14	0.14	5	11	15	21	26	36	55	70	70	70	3	9	13	19	24	34	53	70	70	70
011, 029	X	X	X	L2	1.0	0.70	0.14	0.14	5	11	15	21	26	36	55	70	70	70	3	9	13	19	24	34	53	70	70	70
012, 030	X	X	X	π	1.0	1.4	0.14	0.14	---	---	20	46	57	75	80	80	80	80	---	---	18	44	55	73	78	80	80	80
013, 031	X	X	X	L1	3.0	0.70	0.15	0.05	5	10	14	20	24	31	45	70	70	70	3	8	12	18	22	29	43	70	70	70
014, 032	X	X	X	L2	3.0	0.70	0.15	0.05	5	10	14	20	24	31	45	70	70	70	3	8	12	18	22	29	43	70	70	70
015, 033	X	X	X	π	3.0	1.4	0.15	0.05	---	---	---	17	36	51	80	80	80	80	---	---	---	15	34	49	78	80	80	80
016, 034	X	X	X	L1	5.0	0.70	0.075	0.015	---	---	---	14	17	24	36	60	70	70	---	---	---	12	15	22	34	60	70	70
017, 035	X	X	X	L2	5.0	0.70	0.075	0.015	---	---	---	14	17	24	36	60	70	70	---	---	---	12	15	22	34	60	70	70
018, 036	X	X	X	π	5.0	1.4	0.075	0.015	---	---	---	---	16	38	75	80	80	80	---	---	---	---	14	36	73	80	80	80

1/ Insertion loss measurements shall be made under full load over the frequency range of 100 kHz to 10 MHz. Insertion loss measurements above and below this frequency range shall be under no load.
2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

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Class A filters:

Qualification inspection: Initial qualification may be granted based on qualification to class B. Baseline documentation in accordance with MIL-F-28861 is required and shall be submitted to the qualifying activity prior to completion of quality conformance inspection on the initial lot. Retention of qualification shall be in accordance with class S requirements.

In-process inspection: In accordance with class S requirements, except that the requirements of MIL-C-123 shall not apply.

Quality conformance inspection: The inspection lot definition and inspection of product for delivery shall be in accordance with the class S requirements.

Product assurance level: In accordance with table I.

Part or identifying number: The part or identifying number shall be as follows:

M28861/04 - XXX X X

- XXX: Dash no. per figure 1 and table I
- X: Case finish
- X: Product assurance level per MIL-F-28861 and table I

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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